ZeroWatt Wind

HARVEST THE WIND.
DELIVER THE MOTION.

A WIND-DRIVEN MECHANICAL CONCEPT
FOR URBAN VENTILATION AND POND AERATION

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Towards a Just, Green and Inclusive Energy Transition

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PROBLEM STATEMENT

Airtight urban buildings depend entirely on electric HVAC, which fails whenever load shedding hits. Without airflow, indoor CO_2 and particulate levels climb, and Harvard's COGfx study linked a 500 ppm rise in CO_2 to a 15% drop in cognitive performance. The same neglect shows up outdoors. Stagnant urban ponds lose dissolved oxygen, which leads to algal blooms, foul odor and mosquito breeding. Conventional electric aerators need underwater cabling and add a constant load to the grid.

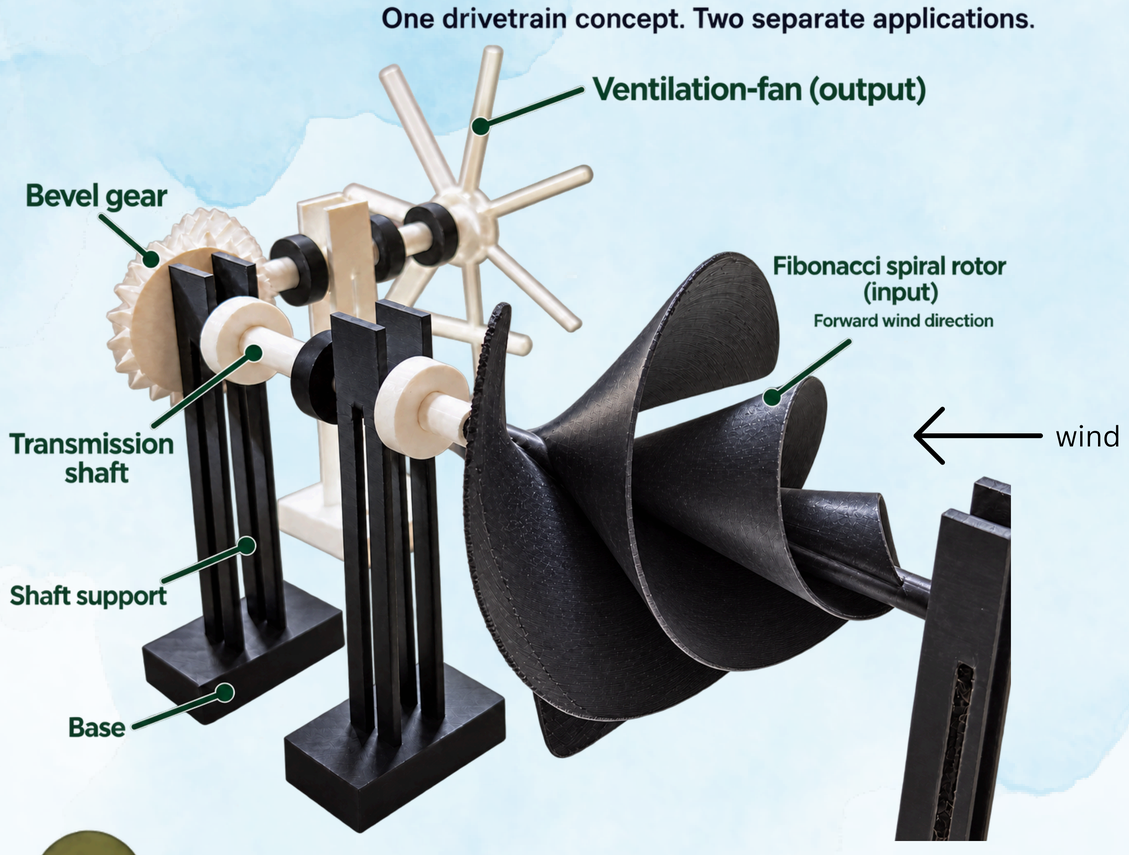
OBJECTIVE

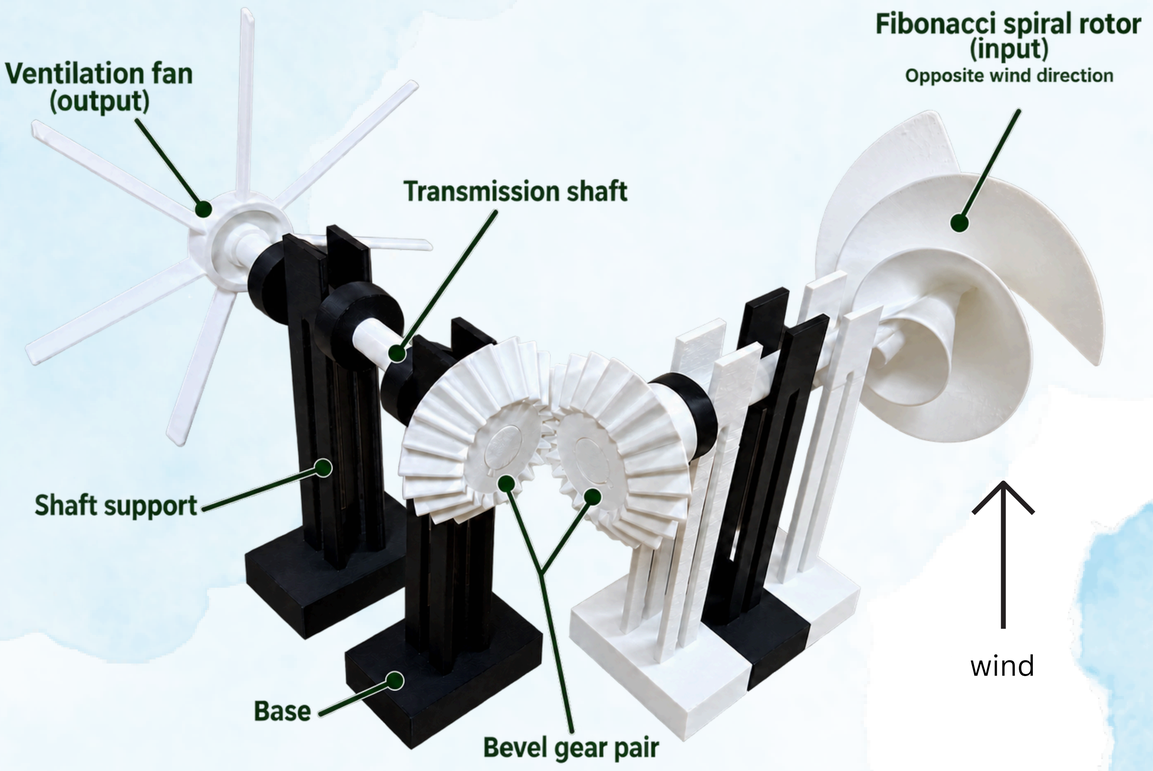
To design and fabricate a passive, wind-driven mechanical prototype that uses a spiral rotor, shafts and bevel gears to power either a ventilation fan or a surface paddle wheel, enabling air exchange in small enclosed spaces or surface mixing in urban ponds without electrical input, with affordability as a design goal.

HOW IT WORKS



One drivetrain concept. Two separate applications.





PROTOTYPE AND FABRICATION

The prototype was designed in SolidWorks and 3D-printed in PLA. Airfoil geometry was generated using Python and SciPy.

Starting wind speed, shaft torque, airflow, drivetrain efficiency and oxygen-transfer performance remain to be measured.

01 | PASSIVE VENTILATION

Designed to exhaust warm air and draw fresh air into occupied spaces.

NACA 65-810 fan blades exhaust hot air pooling near the ceiling. Resulting negative pressure draws fresh air into the occupied zone and removes humidity, reducing mould risk.

02 | POND AERATION

Designed to agitate the water surface and promote air-water exchange.

A surface paddle wheel increases air-water exchange and raises dissolved oxygen. Intermittent, gust-driven operation is intended to disrupt thermal stratification that causes stagnation.

DESIGN FEATURES

- Fibonacci spiral rotor for wind-energy capture.
- Shafts and bevel gears for mechanical transmission through 90°.
- NACA 65-810 fan profile selected for the ventilation design.
- PLA components for prototyping and design iteration

INTENDED IMPACT

- Operation during power outages when sufficient wind is available.
- No electrical input during wind-driven operation.
- Designed for affordability, suited to dense urban and low-income communities
- Improved indoor air quality and healthier urban water bodies
- Potential for local fabrication and repair.